"Are these autofluorescences in the room with us right now?" Quantifying impact of autofluorescence variation on unmixing

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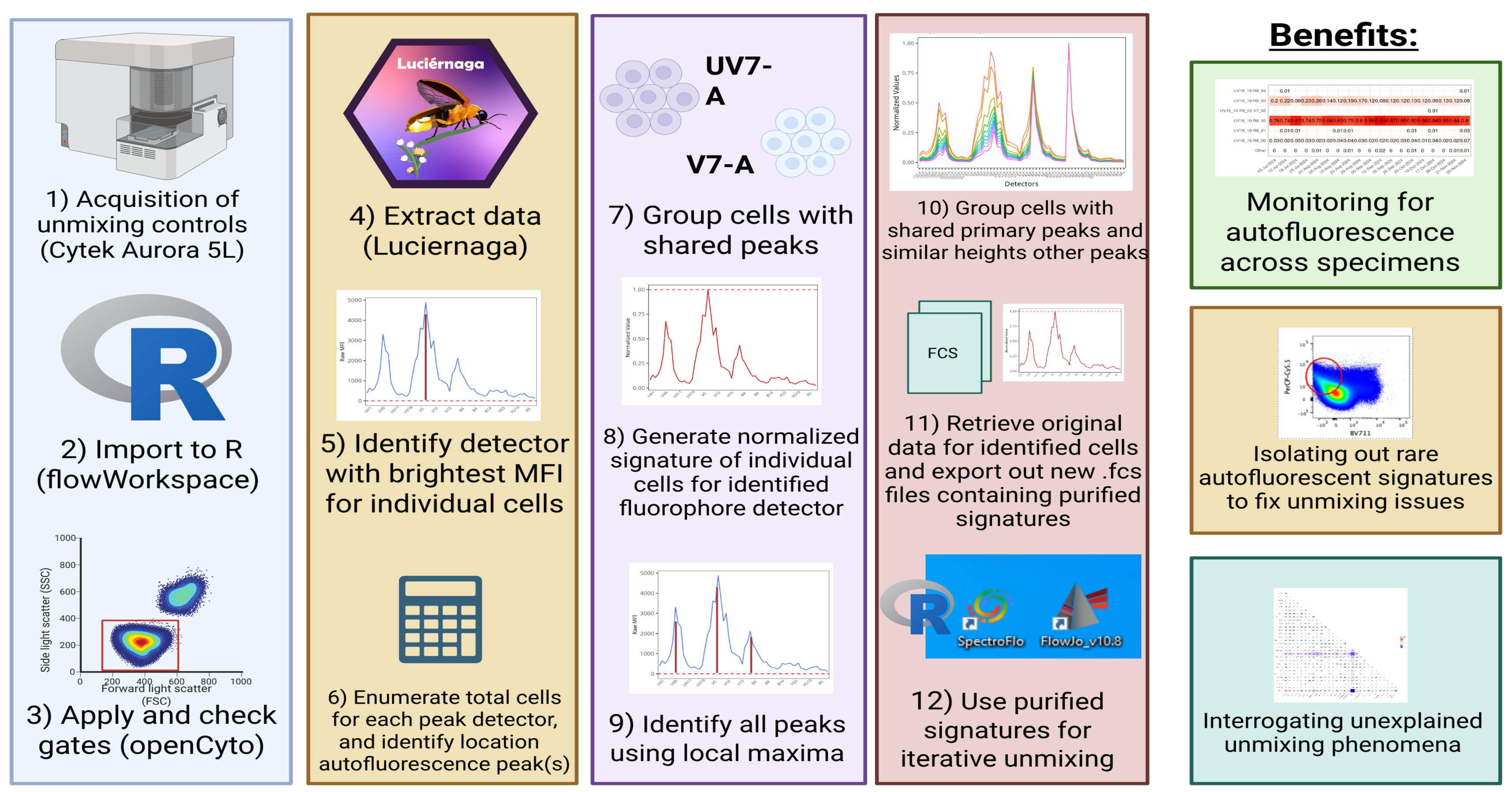
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Introduction

Proper unmixing controls (both single-color and unstained) are critical for successful resolution of similar fluorophores in spectral flow cytometry (SFC). When using cells in the place of beads, the unstained unmixing controls are particularly important, given that they enable subtraction of the autofluorescence background present within single-color unmixing controls, and potentially serve as an additional fluorophore. Autofluorescence, especially in context of human cord and peripheral mononuclear cells (CBMCs and PBMCs), is often treated as a single fluorophore, primarily differing between cell populations in brightness. However, the sources of autofluorescence within a cell may vary and are affected by cell activation, cryopreservation, and fixation during processing. Thus, at the single cell level, autofluorescence signatures may be slightly different. The point at which variation in autofluorescence at the individual cell level goes from being negligible to impacting the resolution of a complex panel is still being addressed. If multiple autofluorescence signatures are present in a population but unaccounted for in the unmixing matrix, uncertainty is introduced, reducing the ability to resolve other fluorophores. This is particularly problematic within large spectral panels with closely related fluorophores and complex marker co-expression patterns. However, the addition of multiple highly similar autofluorescence signatures to an unmixing matrix can increase the complexity with further loss of resolution. We set out to quantitatively interrogate at which point differences in autofluorescence signatures within human mononuclear cells begin to impact panel resolution.

Methods


1) Acquisition of unmixing controls (Cytek Aurora 5L)
2) Import to R (flowWorkspace)
3) Apply and check gates (openCyto)
4) Extract data (Luciernaga)
5) Identify detector with brightest MFI for individual cells
6) Enumerate total cells for each peak detector, and identify location autofluorescence peak(s)
7) Group cells with shared peaks
8) Generate normalized signature of individual cells for identified fluorophore detector
9) Identify all peaks using local maxima
10) Group cells with shared primary peaks and similar heights other peaks
11) Retrieve original data for identified cells and export out new .fcs files containing purified signatures
12) Use purified signatures for iterative unmixing
Benefits:
Monitoring for autofluorescence across specimens
Isolating out rare autofluorescent signatures to fix unmixing issues
Interrogating unexplained unmixing phenomena

Results

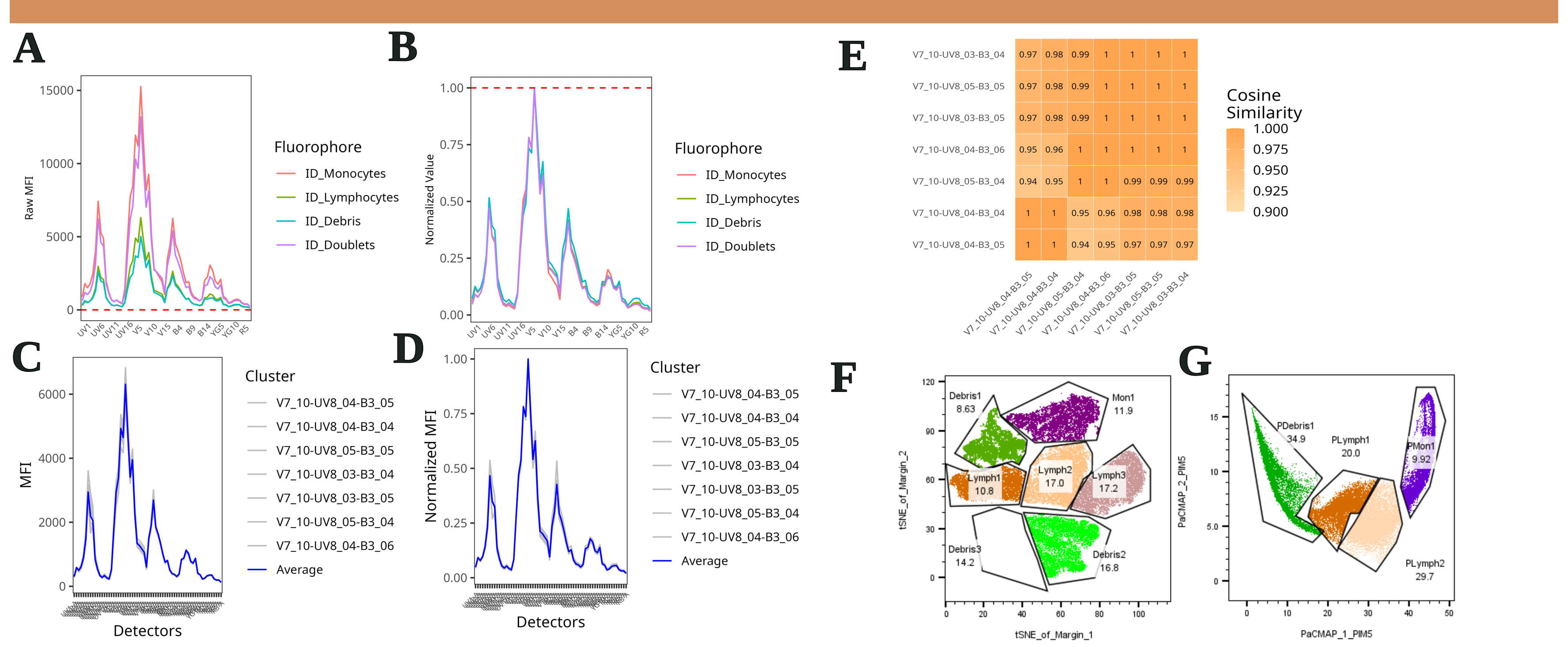


Figure 1. What we expected: "Similar signatures, but different brightness" **A)** The lineplot displays the median fluorescent intensity for the respective cell population. **B)** The lineplot shows the normalized autofluorescence signatures of the different cell populations. **C)** show the median of the MFI for cells within the Luciernaga isolated signature groups. **D)** shows the median of the normalized signature. **E)** The plot shows the cosine values across the different signatures. In **F)** and **G)** dimensionality visualization of the unstained sample was carried out using the MFI values of all detectors, for tSNE and PaCMAP respectively.

Results

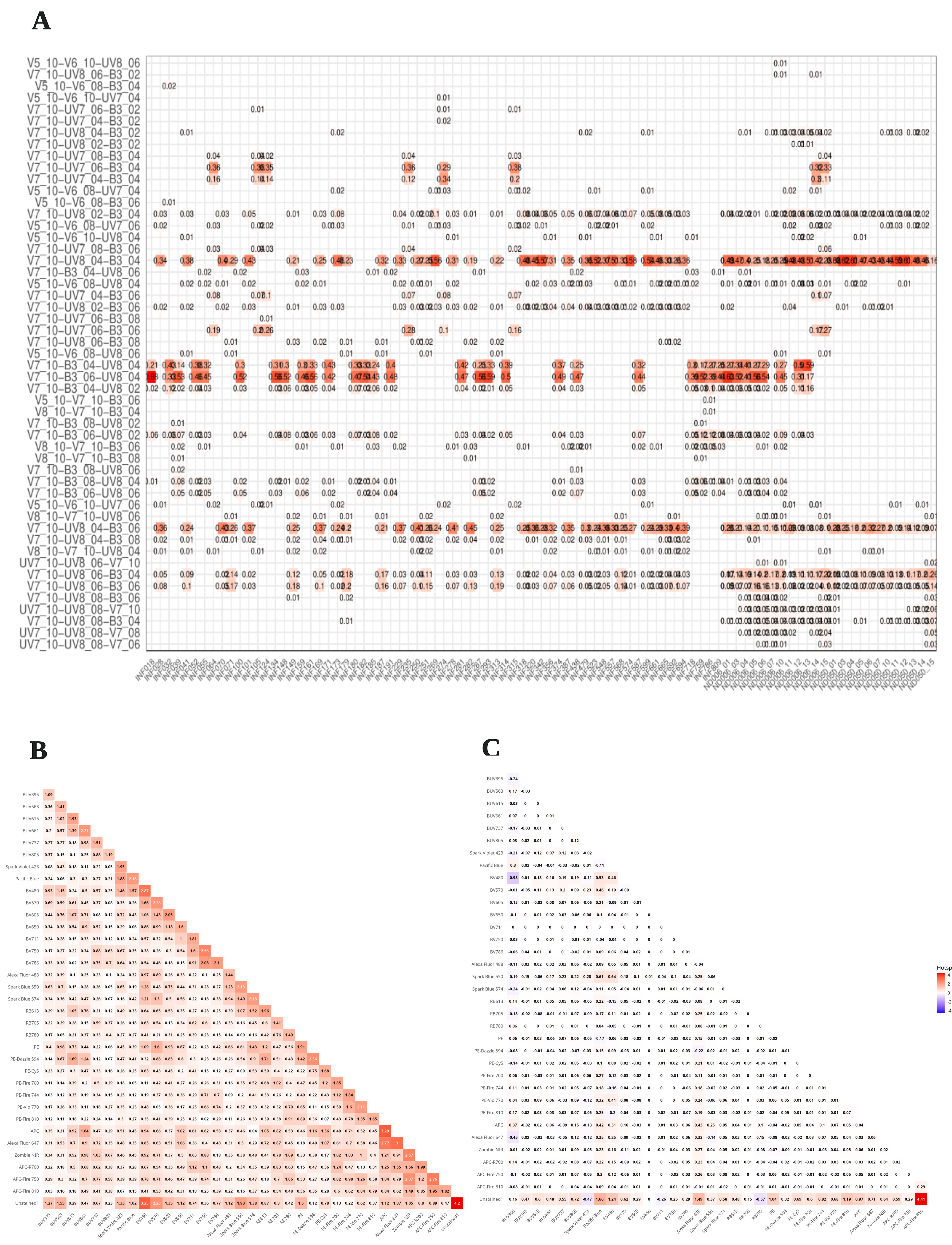


Figure 2. Profiling autofluorescent signatures in Cord Blood Mononuclear Cells in a cohort of specimens Cryopreserved CBMCs were thawed, counted and resuspended at 3x10^6 cells/mL in complete R10 with PMA-ionomycin and.or golgi transport inhibitors for six hours. Between 0.3 - 1x10^6 cells per specimen were aliquoted and processed in the same manner as the full-stained samples. Briefly, cells were washed 5%FBS-PBS, fixed with BD FixPerm, and resuspended in 0.4% PFA-PBS. All samples acquired following day on a 5 laser Cytek Aurora. **A)** The heatmap displays individual specimens on the x-axis, isolated signatures on the y-axis. The ratio of cells for a given signature compared to all other cells from that specimen is shown and visualized in red. **B)** shows a unmixing-dependent spreading (UDS) hotspot matrix for the full-stain panel, when the general autofluorescence signature is incoroporated. **C)** displays the difference between a UDS hotspot matrix for a 34-color panel containing a single unstained signature from that of an identical matrix containing the two separate autofluorescent signatures used for unmixing in Figure 3.

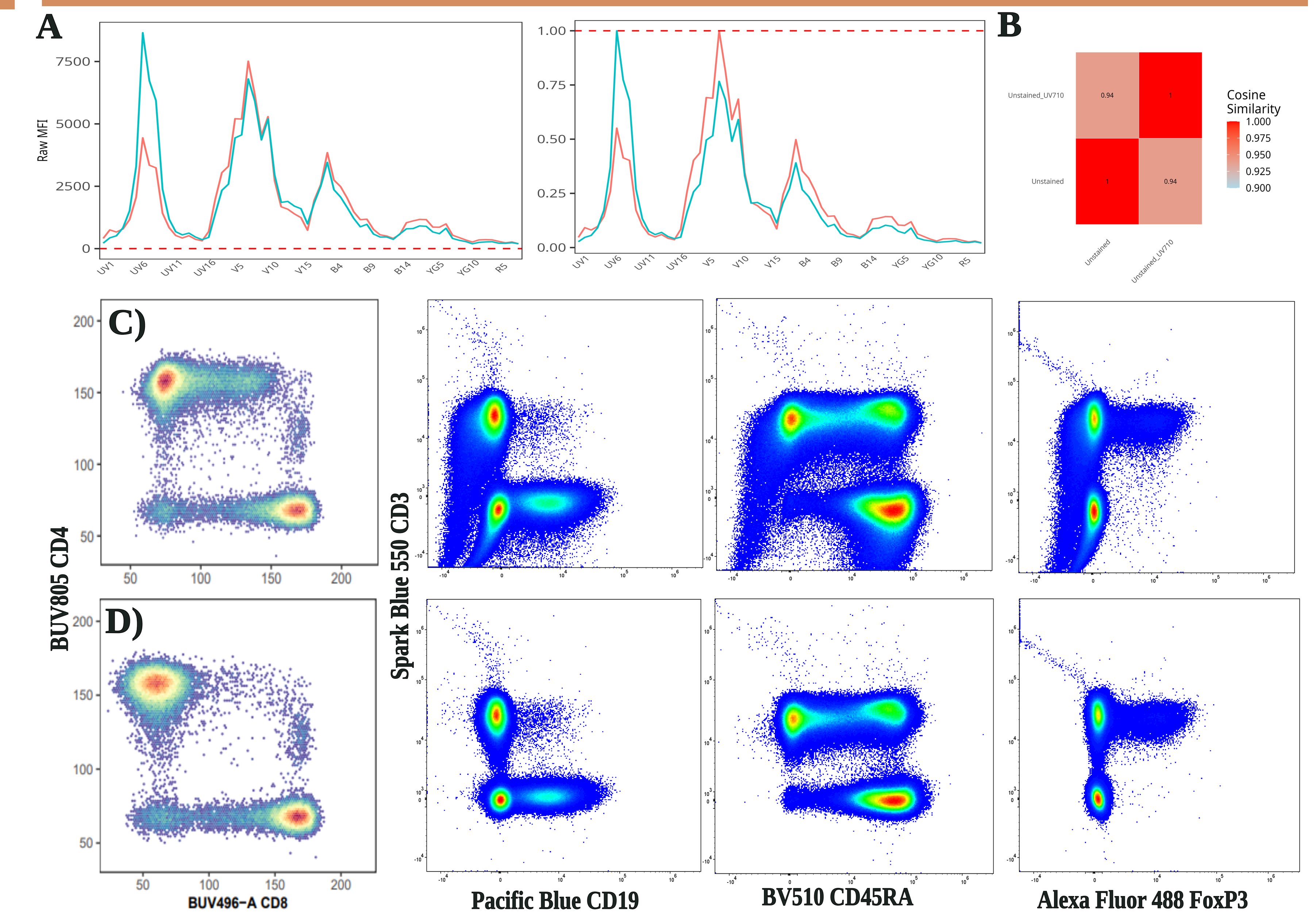
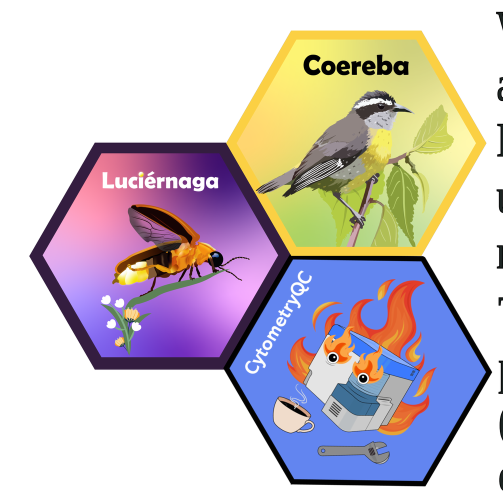

BUV805 CD4
BUV496-A CD8
Spark Blue 550 CD3
Pacific Blue CD19
BV510 CD45RA
Alexa Fluor 488 FoxP3

Figure 3. Rare autofluorescence signatures (<1%) not matching main autofluorescence peak detector sufficient to trigger visible unmixing errors. **A)** The lineplot shows the isolated UV8 autofluorescence signature compared to the main autofluorescence signature. **B)** The plot shows the cosine value of the two signatures. **C)** shows the full-stained sample unmixed with only the main autofluorescence signature, while **D)** shows when an additional UV8 tag is added.

Takeaways

.Lymphocytes and Monocytes typically have similar autofluorescences with each other, mainly distinguished by differences in the MFI intensity. We profiled autofluorescent signatures of individual cells for all individuals in a cohort, and found that most individuals shared the most abundant of the signatures. When we iteratively unmixed these similar signatures that shared the same detector peak with each other, we did not observe any significant unmxing issues. We additionally found rarer autofluorescences present in lymphocytes of some individuals, that differed in their primary peak detector, that substantually affected unmixing, enphasizing collecting enough events to screen and acquire the signatures needed to fix the unmixing

Data availability


Coereba
Luciérnaga
CytometryQC

Would you like to do this your own dataset? The R code used for the analysis and figure generation is available on our GitHub (https://github.com/DavidRach/Autofluorescence_Cyto2025). The single color and unstained unmixing controls will be accessible via ImmPort (SDY3080) in the next release cycle.

The Luciernaga R package (and two other spectral flow cytometry focused projects) are in beta development and available via GitHub (https://github.com/DavidRach/Luciernaga). We are currently editing the documentation and plan to submit them to Bioconductor by the fall.

Take some stickers and track our progress!

Acknowledgements

UMGCCC Flow Cytometry Shared Service U01 HD092308

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